BADR UNIVERSITY IN CAIRO (BUC) — FACULTY OF ENGINEERING

Fluid Power Systems (MEC 431 / MEC 451 / MEC 457) — Quizzes Master Solution & Teaching Guide

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Part 1: Solution of Quiz (2) — Chapter 2: Hydraulic Oils & Theoretical Background

Quiz Question:
"Derive an expression for the pressure and power losses in a hydraulic transmission line of constant diameter in the case of laminar flow."

1.1 Step-by-Step Mathematical Derivation

Consider a straight, rigid, horizontal hydraulic transmission line with inside diameter D, length L, carrying a fluid of density ρ and dynamic viscosity μ at a volumetric flow rate Q.

Step 1: The Darcy-Weisbach General Head Loss Formulation:
The frictional pressure drop ΔP across a pipe of length L is given by:

$$\Delta P = \lambda \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2} \qquad \text{--- Eq. (1)}$$

where: λ = Darcy friction factor, v = mean fluid flow velocity [m/s].

Step 2: Friction Factor for Laminar Flow:
For steady, fully developed laminar flow ($Re < 2000$), analytical integration of the Navier-Stokes equations (or the momentum balance on a fluid cylinder) yields:

$$\lambda = \frac{64}{Re} \qquad \text{--- Eq. (2)}$$

where the Reynolds number is defined as:

$$Re = \frac{\rho v D}{\mu} = \frac{v D}{\nu} \qquad \text{--- Eq. (3)}$$

Step 3: Relationship between Velocity and Flow Rate:
From the continuity equation, mean velocity v in terms of volumetric flow rate Q is:

$$v = \frac{Q}{A} = \frac{Q}{\frac{\pi}{4}D^2} = \frac{4Q}{\pi D^2} \qquad \text{--- Eq. (4)}$$

Step 4: Algebraic Substitution to Derive Pressure Loss:
Substitute Eq. (2) and Eq. (3) into Eq. (1):

$$\Delta P = \left(\frac{64}{\frac{\rho v D}{\mu}} \right) \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2} = \frac{64\mu}{\rho v D} \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2} = \frac{32 \mu L v}{D^2}$$

Now substitute the mean velocity $v = \frac{4Q}{\pi D^2}$:

$$\Delta P = \frac{32\mu L}{D^2} \cdot \left(\frac{4Q}{\pi D^2} \right) = \frac{128\mu LQ}{\pi D^4}$$

Final Pressure Loss Expression (Hagen-Poiseuille Law):

$$\boxed{\Delta P = \frac{128\mu L}{\pi D^4} \cdot Q = R_{hyd} \cdot Q} \qquad [\mathrm{Pa}]$$

where the **Hydraulic Resistance** of the line is defined as:

$$\boxed{R_{hyd} = \frac{128\mu L}{\pi D^4}} \qquad \left[\frac{\mathrm{N \cdot s}}{\mathrm{m}^5} = \frac{\mathrm{Pa \cdot s}}{\mathrm{m}^3} \right]$$

Step 5: Derivation of Power Loss Expression:
Fluid power transmitted is $N = P \cdot Q$. Therefore, the power dissipated across the line due to fluid friction is:

$$\Delta N = Q \cdot P_1 - Q \cdot P_2 = Q \cdot (P_1 - P_2) = Q \cdot \Delta P$$

Substituting the expression for $\Delta P = R_{hyd} \cdot Q$:

$$\Delta N = Q \cdot (R_{hyd} \cdot Q) = R_{hyd} \cdot Q^2$$

Substituting R_{hyd}:

$$\boxed{\Delta N = \frac{128\mu L}{\pi D^4} \cdot Q^2} \qquad [\mathrm{W}]$$

💡 **TA Teaching Tip & Board Notes:**

- **Electrical Analogy:** Emphasize to students that $\Delta P = R_{hyd} \cdot Q$ is the exact fluid equivalent of Ohm's Law $V = R_{el} \cdot I$, and $\Delta N = R_{hyd} \cdot Q^2$ is equivalent to Joule heating $P_{loss} = R_{el} \cdot I^2$.
- **The $1/D^4$ Sensitivity:** Ask students what happens if inner pipe diameter is halved (e.g., from 20 mm to 10 mm). Pressure drop increases by $(2)^4 = 16$ times! This shows why line sizing is paramount.

Part 2: Solution of Quiz — Chapters (1, 2 & 4) [Renewable Energy Dept]

Question 1: Advantages and Disadvantages of Hydraulic Power Systems

Advantages of Hydraulic Systems	Disadvantages of Hydraulic Systems
1. **Extremely High Power-to-Weight Ratio:** Transmits immense forces and torques using compact, lightweight actuators (pumps and cylinders). 2. **Smooth, Stepless Speed Control:** Actuator speed can be adjusted infinitely and smoothly under full load using simple throttling valves. 3. **Instantaneous Reversibility:** Hydraulic actuators can reverse direction almost instantaneously without damage due to low rotating inertia. 4. **Inherent Overload Protection:** Simple pressure relief valves automatically protect all components from mechanical stall or overpressure. 5. **Self-Lubricating:** The working fluid (mineral oil) continuously lubricates internal moving metal surfaces, minimizing wear. 6. **Flexible Power Routing:** Power is transmitted around sharp corners and across long distances via flexible hoses.	1. **Oil Leakage Hazard:** External leaks cause environmental contamination, slip hazards, and fire danger near hot equipment. 2. **Contamination Sensitivity:** Microscopic solid particles cause rapid wear, spool jamming, and valve siltation (demands fine filtration down to $10\ \mu\text{m}$). 3. **Temperature Dependency:** Viscosity changes drastically with temperature, altering line losses, leakage, and cycle times. 4. **Lower Energy Efficiency:** Conversion losses (mechanical → hydraulic → mechanical) and throttling drops limit overall efficiency (typically 70–85%). 5. **Cavitation & Air Entrainment:** Low suction pressures cause vapor cavities to implode violently, causing erosion and severe noise.

Question 2: Viscous Friction and Damping Effect on a Moving Spool

Problem Statement: Moving parts in oil experience viscous drag due to fluid shear stress. A cylindrical spool moves axially with velocity v inside a sleeve with radial clearance c (2 to 10 µm). Derive an expression for the friction force resisting the moving spool.

Step 1: Shear Stress from Newton's Viscosity Law:
Because radial clearance c is extremely small ($2 - 10\ \mu\text{m}$), the velocity distribution across the gap can be assumed strictly linear:

$$\frac{du}{dy} = \frac{v}{c}$$

The fluid shear stress τ acting on the outer cylindrical surface of the spool is:

$$\tau = \mu \frac{du}{dy} = \mu \frac{v}{c} \qquad \text{--- Eq. (2.12)}$$

Step 2: Cylindrical Shearing Surface Area:
For a spool land of diameter D and axial length L, the wetted cylindrical contact area is:

$$A_s = \pi D L$$

Step 3: Total Viscous Friction Force:
The total viscous friction force resisting motion is the product of shear stress and area:

$$F_v = \tau \cdot A_s = \left(\mu \frac{v}{c}\right) \cdot (\pi DL) = \left(\frac{\pi \mu DL}{c}\right) \cdot v \qquad \text{--- Eq. (2.13)}$$

Viscous Damping Coefficient:

$$\boxed{F_v = f_v \cdot v} \qquad \text{where} \qquad \boxed{f_v = \frac{\pi \mu DL}{c}} \quad [\text{N} \cdot \text{s/m}]$$

💡 **TA Note:** If the spool has multiple lands (e.g., L_1 and L_2), total length is simply $L = \sum L_i$. Note that f_v is inversely proportional to clearance c: tighter clearance increases viscous damping!

Question 3: Power Transmission & Causality Relation of System Variables

Diagram Overview:
$\text{Electric System } (e, i) \rightarrow \text{Electric Motor } (T, \omega = 2\pi n) \rightarrow \text{Pump } (P, Q) \rightarrow \text{Cylinder } (F, v) \rightarrow \text{Load}$

3.1 Detailed Discussion of Variable Transformations & Equations

1. **Electric System → Electric Motor:**
   Electrical power $N_e = e \cdot i$ (voltage × current) is converted into rotary mechanical power $N_m = T \cdot \omega = 2\pi n T$.
   The motor delivers rotational speed n [rev/s] and shaft torque T [N·m].
2. **Electric Motor → Hydraulic Pump:**
   The positive displacement pump is driven at speed n. Neglecting slip, it delivers a volumetric flow rate:

   $$Q = V_g \cdot n \qquad [V_g = \text{geometric displacement per revolution, } \text{m}^3/\text{rev}]$$

   The pressure P created by load resistance acts back on the pump chambers, generating a resisting load torque:

   $$T = \frac{P \cdot V_g}{2\pi} \qquad [\text{Torque demanded by pump from motor}]$$

3. **Hydraulic Pump → Hydraulic Cylinder (Actuator):**
   The fluid flow Q enters the cylinder chamber of area A_p, driving the piston at linear velocity:

   $$v = \frac{Q}{A_p}$$

   The piston pushes against the external load resistance force F. The fluid pressure required is established as:

   $$P = \frac{F}{A_p}$$

3.2 The Fundamental Concept of System Causality

Forward (Kinematic) Causality:
Rotational speed of prime mover (n) → creates pump flow rate (Q) → determines linear piston velocity (v).
Speed drives flow; flow drives velocity!

Backward (Dynamic) Causality:
External load resistance establishes force (F) → creates hydraulic pressure ($P = F/A_p$) → creates pump resisting

torque ($T = PV_g/2\pi$) → draws electric current (i) from the grid.
Load dictates force; force creates pressure; pressure demands torque; torque demands electric current!

> **Golden Rule of Fluid Power:** *"The pump does NOT create pressure; the pump creates FLOW. Pressure is caused ONLY by the resistance to flow!"*

Part 3: Solution of Quiz (5) — Chapter 5: Hydraulic Control Valves

Complete identification of all 13 valves, their official standard ISO 1219 symbols, and operational explanations:

No.	Valve Standard Name	Standard ISO Symbol	Working Principle & Circuit Function
1	**Direct-Operated Pressure Relief Valve** *(Normally Closed)*	**[Square, P→T arrow, internal pilot from P, spring]**	Safety valve connected in parallel with pump. When line pressure exceeds adjustable spring preload, poppet opens and dumps excess flow to tank, limiting maximum system pressure.
2	**Direct-Operated Sequence Valve** *(with External Drain)*	**[Square, P→A, internal pilot from P, external drain T]**	Normally closed valve used to direct flow to a secondary branch only after the primary actuator pressure reaches the predetermined spring setting. External drain prevents backpressure buildup.
3	**2/2 Directional Control Valve (DCV)** *(Spool type, Push-button, Spring-return)*	**2 Ports, 2 Positions [Normal: Blocked | Actuated: P→A]**	Simple On/Off valve. In normal de-energized position, flow is blocked. When button/solenoid is actuated, spool shifts to connect port P directly to port A.
4	**3/2 Directional Control Valve (DCV)** *(Spool type, Spring-return)*	**3 Ports (P, A, T), 2 Positions [Normal: A→T, P blocked | Actuated: P→A]**	Controls single-acting cylinders. In normal position, port A is vented to tank T. In actuated position, port P feeds port A while tank port T is blocked.
5	**4/3 DCV with Tandem Center** *(Solenoid-operated, Spring-centered)*	**3 Positions, 4 Ports (P, T, A, B) Center: P→T, A & B blocked**	In neutral center position, pump delivery is unloaded to tank at 0 bar (saving electrical energy and avoiding heating), while cylinder ports A & B are blocked, locking the actuator in place.
6	**Spring-Loaded Direct-Operated Check Valve**	**Ball & Seat with Spring (Cracking pressure ≈ 0.5–3 bar)**	Allows free flow in forward direction when pressure overcomes spring preload. Completely blocks reverse flow as ball is pressed against conical seat.
7	**Direct-Operated Check Valve without Spring**	**Ball & Seat without Spring (Relies on gravity/flow)**	Free flow in one direction with near-zero opening pressure. Closed in reverse by backward flow and gravity. Used strictly in vertical orientations.
8	**Direct-Operated In-Line Poppet Check Valve**	**Construction Cutaway: Poppet, Guide Sleeve, Return Spring**	Industrial cartridge/in-line poppet construction showing conical guided poppet with internal helical compression spring and radial guide lands.
9	**Adjustable Throttle Valve (Restrictor)**	**Two brackets with transverse 45° arrow**	Bi-directional variable restriction. Flow rate is proportional to $\sqrt{\Delta P}$ and fluid viscosity. Acts as basic speed control device.
10	**Throttle-Check Valve** *(One-Way Flow Control Valve)*	**Adjustable throttle in parallel with reverse check valve**	Controls speed in one direction (meter-in or meter-out), while permitting free unrestricted return flow in the opposite direction through the bypass check valve.

11	**Fixed Throttle & Viscosity-Independent Orifice**	**Fixed restrictor & Sharp-edged orifice symbol**	Sharp-edged orifice has discharge coefficient $C_d \approx 0.611$ that remains nearly constant despite wide oil temperature and viscosity changes.
12	**2-Way Pressure-Compensated Flow Control Valve**	**Throttle + Upstream/Downstream reducing spool in series**	Maintains constant flow rate despite fluctuating actuator load pressures. Compensator spool automatically modulates its opening to keep pressure drop across throttle constant.
13	**3-Way Pressure-Compensated Flow Control Valve** *(Bypass / Priority Type)*	**Throttle + Bypass relief spool connected to tank T**	Delivers constant regulated flow to primary actuator while diverting all excess pump flow directly to tank or secondary circuit at load pressure (energy-saving).

Part 4: Solution of Quiz (6) — Chapter 6: Hydraulic Accessories

Question 1: Sizing of a Hydraulic Accumulator (Adiabatic Process)

Problem Statement:
Calculate the size (V_0) of a hydraulic accumulator necessary to deliver five liters of oil ($\Delta V = 5\ \text{L}$) between operating pressures of $p_2 = 200\ \text{bar}$ and $p_1 = 100\ \text{bar}$, if the precharge pressure is $p_0 = 90\ \text{bar}$ (gauge pressure), assuming an adiabatic compression/expansion process ($\kappa = 1.4$).

Step-by-Step Mathematical Solution:

Step 1: Convert Given Pressures to Absolute Values:
Gas law calculations require absolute thermodynamic pressures ($P_{abs} = P_{gauge} + 1.013\ \text{bar} \approx P_{gauge} + 1\ \text{bar}$):

$$p_0 = 90 + 1 = 91\ \text{bar (abs)} = 9.1 \times 10^6\ \text{Pa}$$

$$p_1 = 100 + 1 = 101\ \text{bar (abs)} = 10.1 \times 10^6\ \text{Pa}$$

$$p_2 = 200 + 1 = 201\ \text{bar (abs)} = 20.1 \times 10^6\ \text{Pa}$$

$$\Delta V = 5.0\ \text{L} = 5.0 \times 10^{-3}\ \text{m}^3$$

Step 2: Adiabatic Expansion Governing Equation:
For nitrogen gas undergoing rapid expansion during oil discharge, heat transfer with the environment is negligible. The process is governed by the adiabatic law:

$$p \cdot V^{\kappa} = \text{constant} \qquad \text{where for Diatomic Nitrogen, } \kappa = 1.4$$

Step 3: Gas Volume at Pressures p_1 and p_2:
From state 0 (precharge: p_0, V_0) to state 1 (minimum working pressure: p_1, V_1):

$$p_0 V_0^{\kappa} = p_1 V_1^{\kappa} \implies V_1 = V_0 \left(\frac{p_0}{p_1}\right)^{\frac{1}{\kappa}}$$

From state 0 to state 2 (maximum working pressure: p_2, V_2):

$$p_0 V_0^{\kappa} = p_2 V_2^{\kappa} \implies V_2 = V_0 \left(\frac{p_0}{p_2}\right)^{\frac{1}{\kappa}}$$

Step 4: Relate Gas Volumes to Oil Delivered (ΔV):
The volume of oil delivered to the system as pressure drops from p_2 to p_1 equals the gas expansion volume:

$$\Delta V = V_1 - V_2 = V_0 \left[\left(\frac{p_0}{p_1}\right)^{\frac{1}{\kappa}} - \left(\frac{p_0}{p_2}\right)^{\frac{1}{\kappa}}\right]$$

Step 5: Compute Numerical Value of Accumulator Size V_0:

$$\frac{1}{\kappa} = \frac{1}{1.4} = 0.714286$$

$$\left(\frac{p_0}{p_1}\right)^{\frac{1}{\kappa}} = \left(\frac{91}{101}\right)^{0.714286} = (0.90099)^{0.714286} = 0.92790$$

$$\left(\frac{p_0}{p_2}\right)^{\frac{1}{\kappa}} = \left(\frac{91}{201}\right)^{0.714286} = (0.45274)^{0.714286} = 0.56743$$

$$\text{Bracket term} = 0.92790 - 0.56743 = 0.36047$$

$$V_0 = \frac{\Delta V}{0.36047} = \frac{5.0\ \text{L}}{0.36047} = \mathbf{13.87\ Liters}$$

Final Calculated Size of Accumulator:

Nominal Gas Capacity: **V_0 = 13.87 Liters**

*(Engineering Selection: Standard commercial accumulator shell of **16 Liters** or **20 Liters** would be selected from manufacturer catalogs).*

Question 2: Principles of Operation & Applications of Hydraulic Accumulators

2.1 Principle of Operation

Liquids are virtually incompressible and cannot store potential energy in small volumes. In an **oleo-pneumatic accumulator**, an inert compressed gas (dry Nitrogen, N_2) is sealed inside a rigid pressure vessel alongside the hydraulic oil, separated by a flexible bladder, piston, or diaphragm. Because gas is highly compressible, fluid forced into the vessel compresses the gas, storing potential energy. When line pressure drops, the expanding gas discharges the stored oil back into the circuit.

2.2 Main Industrial Applications

1. **Auxiliary Energy Storage (Peak Shaving):** Delivers intermittent high flow during short cycle phases, allowing the pump and motor to be sized for the *average* rather than *peak* flow rate (saves initial equipment cost and running power).
2. **Emergency Power Source (Fail-Safe):** Stores sufficient fluid to safely complete an operation (e.g., closing a dam sluice gate, raising aircraft landing gear, or clamping a heavy chuck) if electrical power to the pump fails.
3. **Leakage and Pressure Compensation:** Maintains system clamping pressure over long holding cycles while the pump is completely shut off or unloaded to tank.
4. **Hydraulic Shock & Water Hammer Suppression:** Absorbs destructive pressure spikes caused by sudden valve closure or actuator deceleration.
5. **Pulsation Dampening:** Smooths out flow and pressure ripple generated by positive displacement pumps (gear and piston pumps).